Post-Quantum Cryptography Migration

Reading Guide

Compiled by the Spaced Research Ledger

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Post-Quantum Cryptography Migration is a survey of one large, slow-moving event: the replacement of the public-key cryptography that secures nearly all digital communication. Quantum computers, once large enough, will break the algorithms in use today. Standards bodies have chosen replacements, and deploying them now touches every layer of computing, from silicon to government policy.

This collection covers that work in nineteen parts. The parts are ordered by dependency: hardware before the software built on it, software before the systems deployed on it, deployments before the rules that govern them. Read front to back, each part builds on the ones before. Read alone, every part stands on its own.

The threat driving all of it has a name: harvest now, decrypt later. Encrypted traffic recorded today can be decrypted once a capable quantum computer exists, so anything meant to stay secret for years is already at risk. That is why key exchange has migrated fast across browsers, clouds, and messengers, while signatures, which only matter once such a computer is live, move on a slower clock.

Every statement in every part carries a numbered citation to a public source, listed at the end of that part. When sources disagree, the disagreement is reported under its own heading rather than resolved by guesswork. Each part states when it was last updated.

How to Read This Collection

Each part opens with a summary written for someone who reads nothing else. Numbered sections follow, one per subtopic, and each section starts with a short italic paragraph explaining in plain language what the thing is and why it matters. Inside each section, Recent Developments covers what changed lately and Current State covers what is durably true. Unresolved Conflicts, where present, records disagreements between sources with a note on what would settle each one.

Start with the summary of any part that touches your work. The intro of each part names its sections by number, so you can jump straight to the subtopic you need. The parts most readers want first are Part 6 for the web's protocols, Part 7 for certificates and identity, and Part 19 for the deadlines.

The Parts

Silicon & Firmware

Part 1 - Hardware Acceleration & Silicon. How fast the new algorithms run on real processors, the accelerator cards built for heavy loads, and the licensable chip designs putting the algorithms into future silicon.

Part 2 - Firmware & Root of Trust. The boot chains, TPM chips, and firmware-update signing that decide what a computer trusts before it even starts, carrying the earliest deadlines in the whole migration.

Operating Systems

Part 3 - Operating Systems & Endpoints. Windows, Apple, Linux, Android, mainframes, and embedded systems: who ships the algorithms, whose defaults are on, and where the gaps sit.

Cryptographic Foundations

Part 4 - Crypto Libraries & Providers. OpenSSL and its forks, the language runtimes, the reference implementations, and the first real attacks on post-quantum code.

Part 5 - HSMs, KMS & Key Management. The hardware modules and key managers that guard an organization's most sensitive keys, the validation pipeline that gates their purchase, and the archives exposed to decades of decryption risk.

Protocols & Trust

Part 6 - Network & Transport Protocols. TLS, QUIC, VPNs, SSH, Wi-Fi's cousins on the wire, DNS, and routing: where hybrid key exchange crossed from experiment to majority.

Part 7 - PKI, Certificates, Identity & Authentication. The certificate authorities, enrollment protocols, identity providers, and passkeys, where zero post-quantum roots exist in any browser trust store and Merkle Tree Certificates are the chosen way out.

Building & Shipping Software

Part 8 - Application Dev Stacks & Languages. Web servers, language runtimes, database drivers, and browsers, the layer that inherits its cryptography from below and passes it to everything above.

Part 9 - Software Supply Chain. Package signing, code signing, and build provenance: the first hybrid-signed Linux packages and Android's hybrid app signatures.

Part 10 - Containers, Orchestration, Service Mesh. Kubernetes and its neighbors, quantum-safe on the wire by inheritance and classical in every certificate they issue.

Where It Runs

Part 11 - Cloud Infrastructure. AWS, Azure, Google Cloud, the other providers, the CDNs that finished first, and the attestation chains that have not started.

Part 12 - Security & Network Appliances. Firewalls, load balancers, routers, and encryptors, plus the middleboxes that silently drop the bigger handshakes.

Part 13 - Enterprise Infrastructure & Storage. Hypervisors, server management chips, backup systems, and the tape drive that beat everyone to post-quantum readiness.

Part 14 - Enterprise Applications & Delivery. ERP platforms, video conferencing, and remote desktops: the last stop of the migration, with published vendor clocks.

Part 15 - Email & Messaging. OpenPGP's new standard, S/MIME's machinery, and the secure messengers where post-quantum protection is furthest along anywhere.

Edges & Sectors

Part 16 - Enterprise IoT & OT. The grid, factories, cars, medical devices, payment terminals, satellites, and the small radios where post-quantum keys barely fit.

Part 17 - Financial Messaging & Sector Rails. SWIFT, settlement systems, and card networks, including the first live post-quantum tests inside a central-bank payment platform.

Finding & Governing It

Part 18 - Developer & Migration Tooling. The inventories, scanners, validators, and side-channel research that make migration manageable, led by the cryptographic bill of materials.

Part 19 - Governance, Mandates & Timelines. The deadlines: executive orders, European roadmaps, certification cutoffs, and the procurement language reaching into contracts.

About the Project

The Spaced Research Ledger is an independent, not-for-profit research project designed and maintained by Ridley DiSiena. It is a subject-neutral method. The automated pipeline can be pointed at any narrow field, gather what public sources currently say about it, and keep that picture current as the field moves. This collection is that method applied to the migration to post-quantum cryptography.

The Ledger works the same way on every subject. It gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it.

These reports state what their sources say, with every statement cited. They take no editorial position, represent no government, company, or other organization, and endorse no product. The project has no sponsor and no commercial interest in anything it covers. It is published for anyone who finds it useful.